

Testing Invariance and Generalization of Geometry-Aware Sharpness Minimization Methods

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1 Motivation

Recent advances in deep learning optimization highlight the importance of sharpness-aware methods that seek flat minima to improve generalization. One such method, Sharpness-Aware Minimization (SAM) (Foret et al., 2021), has shown promising results. However, it is sensitive to the choice of model parameterization, as its perturbation radius is defined in the Euclidean metric.

Adaptive SAM (ASAM) (Kwon et al., 2021) and Monge SAM (M-SAM) (Jacobsen and Arvanitidis, 2025) try to address these limitations by introducing scale invariance and geometry-aware metrics, respectively. However, existing evaluations of M-SAM are limited to a single fine-tuning task on ResNet-18 and a multi-modal alignment task; a systematic comparison of all three optimizers under controlled reparametrization across multiple architectures has not been done.

Understanding how these methods behave under reparametrization is crucial for developing more robust optimization techniques, and this gap motivates our empirical work on benchmarking SAM, ASAM, and M-SAM on ResNet-18 and ViT-B/32 architectures with a focus on reparametrization invariance and generalization performance.

2 Methods

We implement three optimizers from scratch (SAM, ASAM, M-SAM) and compare them under controlled conditions.

SGD serves as the reference baseline: vanilla stochastic gradient descent with momentum and weight decay, trained with a cosine-annealing learning rate schedule.

SAM (Foret et al., 2021) augments SGD with a two-step update. In the first step the parameters are perturbed to the approximate local loss maximum within an ℓ_2 ball of radius ρ :

$$\hat{\epsilon}(w) = \rho \cdot \frac{\nabla_w \mathcal{L}(w)}{\|\nabla_w \mathcal{L}(w)\|_2},$$

and in the second step the base SGD update is applied at the perturbed point:

$$w_{t+1} = w_t - \eta \nabla_w \mathcal{L}(w_t + \hat{\epsilon}(w_t)).$$

Because the perturbation is defined in Euclidean space it is *not* invariant to function-preserving rescalings of the weights.

ASAM (Kwon et al., 2021) addresses this by replacing the ℓ_2 norm with an adaptive (parameter-scaled) norm. The perturbation becomes

$$\hat{\epsilon}_{\text{ASAM}}(w) = \rho \cdot \frac{T_w^2 \nabla_w \mathcal{L}(w)}{\|T_w \nabla_w \mathcal{L}(w)\|_2},$$

where $T_w = \text{diag}(|w| + \eta)$ with a small floor $\eta = 0.01$, and the weight update is:

$$w_{t+1} = w_t - \eta \nabla_w \mathcal{L}(w_t + \hat{\epsilon}_{\text{ASAM}}(w_t)).$$

Multiplying the perturbation by $|w|^2$ makes it scale-invariant: if every weight is multiplied by a scalar α , $\hat{\epsilon}_{\text{ASAM}}$ is unchanged. Our implementation exactly follows this formula.

M-SAM (Jacobsen and Arvanitidis, 2025) takes a Riemannian geometry perspective. The loss is viewed as a function on the model-function manifold; the Monge metric on that manifold yields a correction factor $1/\sqrt{1 + \|\nabla_w \mathcal{L}(w)\|_2^2}$ applied to the SAM perturbation:

$$\hat{\epsilon}_{\text{MSAM}}(w) = \frac{\rho}{\|\nabla_w \mathcal{L}(w)\|_2} \cdot \frac{1}{\sqrt{1 + \|\nabla_w \mathcal{L}(w)\|_2^2}} \cdot \nabla_w \mathcal{L}(w),$$

with the weight update:

$$w_{t+1} = w_t - \eta \nabla_w \mathcal{L}(w_t + \hat{\epsilon}_{\text{MSAM}}(w_t)).$$

This factor shrinks large-gradient perturbations, promoting geometry-aware and reparametrization-robust updates.

Model and Dataset. We use **ResNet-18** (He et al., 2016) adapted for CIFAR-10 (Krizhevsky, 2009): the standard ImageNet stem (7×7 conv, stride 2, max-pool) is replaced with a 3×3 stride-1 conv and an identity layer to retain spatial resolution for 32×32 inputs. The final fully-connected layer maps from 512 to 10 classes. All experiments use a single random seed (seed 42) at this stage; multi-seed evaluation is planned for the final report.

Training Protocol. All models are trained for 200 epochs on CIFAR-10 with:

- Batch size 256, learning rate 0.1, momentum 0.9, weight decay 5×10^{-4} ;
- Cosine-annealing LR schedule (no warmup);
- Perturbation radius ρ swept over $\{0.005, 0.01, 0.02, 0.03, 0.05\}$ for SAM and M-SAM; a single $\rho = 0.5$ for ASAM (following the original paper’s recommended scale).

For both SAM variants two forward-backward passes are performed per iteration.

Same protocol will be applied to ViT-B/32 on CIFAR-10 as a stretch goal, pending resolution of the GELU reparametrization issue.

Evaluation Metrics. We track final test accuracy, training accuracy, cross-entropy loss curves, and the *divergence rate* $\Delta = \mathcal{L}_{\text{test}} - \mathcal{L}_{\text{train}}$ as a proxy for generalization gap. For reparametrization invariance we measure the variance of final test accuracy across scaling factors $\alpha \in \{0.1, 0.5, 1.0, 2.0, 10.0\}$ applied to a function-preserving ReLU-layer rescaling. Sharpness will be estimated via the Hutchinson stochastic trace estimator $\text{tr}(H)/d$ with Rademacher vectors, which avoids forming the full Hessian.

3 Preliminary Experiments and Results

Baseline Training. We have completed the full baseline sweep (200 epochs each) for all four optimizer (SGD, SAM, ASAM, M-SAM) families on ResNet-18/CIFAR-10, producing 12 checkpoint files: SGD ($\rho=0$), SAM at five ρ values, ASAM at $\rho=0.5$, and M-SAM at five ρ values. All models reach 100% training accuracy.

Table 1 reports test accuracy and divergence rate Δ for every configuration.

Table 1: Full baseline results on ResNet-18/CIFAR-10 (200 epochs, seed 42). Best row per optimizer in bold.

Optimizer	ρ	Acc (%)	Δ
SGD	0.000	95.46	0.1777
SAM	0.020	95.75	0.1494
SAM	0.005	95.23	0.1784
SAM	0.010	95.31	0.1729
SAM	0.030	95.70	0.1473
SAM	0.050	95.55	0.1407
ASAM	0.500	95.79	0.1440
M-SAM	0.005	95.47	0.1736
M-SAM	0.010	95.50	0.1683
M-SAM	0.020	95.64	0.1559
M-SAM	0.030	95.76	0.1451
M-SAM	0.050	95.75	0.1384

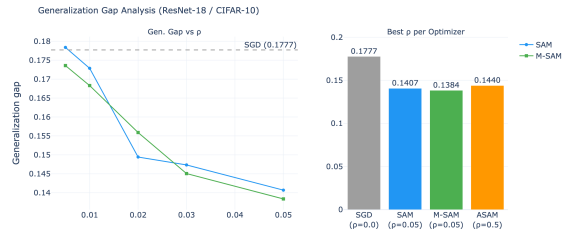


Figure 1: Generalization gap analysis (ResNet-18/CIFAR-10, seed 42). *Left*: divergence rate vs. ρ ; M-SAM consistently lower. *Right*: best- ρ per optimizer vs. SGD (0.1777). M-SAM at $\rho=0.05$ achieves the largest reduction (-22%).

Sharpness and Flatness Analysis. We ran the Hutchinson stochastic trace estimator ($n=20$ Rademacher vectors, max batch 64) on all 12 checkpoints and computed 2D loss landscapes using

two Gram-Schmidt-orthogonalized filter-normalized random directions (31×31 grid, $\alpha, \beta \in [-1, 1]$). Table 2 reports $\text{tr}(H)/d$ for each configuration; Figure 2 shows the 3D surface landscape for the best checkpoint per optimizer.

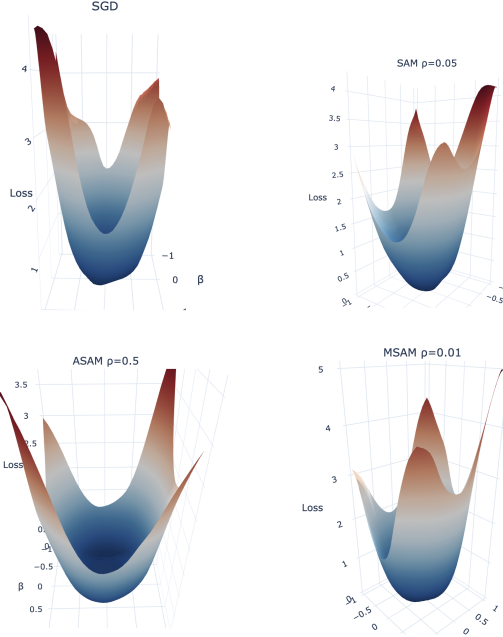


Table 2: Hutchinson $\text{tr}(H)/d$ for all 12 checkpoints. Lower is flatter.

Optimizer	ρ	$\text{tr}(H)/d$
SGD	0.000	6.25×10^{-4}
SAM	0.005	6.35×10^{-4}
SAM	0.010	4.02×10^{-4}
SAM	0.020	4.45×10^{-4}
SAM	0.030	5.16×10^{-4}
SAM	0.050	3.43×10^{-4}
ASAM	0.500	3.17×10^{-4}
M-SAM	0.005	6.24×10^{-4}
M-SAM	0.010	4.29×10^{-4}
M-SAM	0.020	4.40×10^{-4}
M-SAM	0.030	2.61×10^{-4}
M-SAM	0.050	1.89×10^{-4}

Figure 2: 3D filter-normalized loss landscapes for the best checkpoint per optimizer (ResNet-18/CIFAR-10). Filter-normalized directions span a 31×31 grid over $\alpha, \beta \in [-1, 1]$. M-SAM and ASAM show broader, flatter basins.

Several observations stand out. First, SAM at $\rho=0.005$ is *sharper* than SGD (6.35×10^{-4} vs. 6.25×10^{-4}), confirming that an under-sized perturbation radius fails to flatten the loss landscape and can hurt generalization. Second, M-SAM at $\rho=0.05$ achieves the lowest sharpness across all runs (1.89×10^{-4} , approximately 70% flatter than SGD), consistent with its best divergence rate. Third, ASAM (3.17×10^{-4}) is flatter than SGD and comparable to the best SAM setting, yet achieves the highest raw test accuracy, suggesting that its adaptive normalization improves generalization through a mechanism not fully captured by the scalar Hessian trace. The full 3D loss landscapes for all 12 checkpoints are shown in Figure 3 (Appendix).

Infrastructure and Implementation Validation. The training pipeline, data loading (HuggingFace CIFAR-10 cached locally), optimizer implementations, and checkpoint saving have all been validated. The Hutchinson sharpness estimator and loss landscape visualizer are also implemented and unit-tested.

The reparametrization utility (`apply_relu_reparam`) for ResNet-18 BasicBlocks has been implemented and verified. It exploits the positive 1-homogeneity of ReLU, i.e. $\text{ReLU}(\alpha x) = \alpha \text{ReLU}(x)$ for $\alpha > 0$, to construct a function-preserving weight scaling. Within each BasicBlock the computation chain is:

$$\text{conv1} \rightarrow \text{BN1} \xrightarrow{\times \alpha} \text{ReLU} \xrightarrow{\times \alpha} \text{conv2} \xrightarrow{\times 1/\alpha}$$

The α factor is injected by scaling BN1’s affine parameters ($\gamma \leftarrow \alpha\gamma, \beta \leftarrow \alpha\beta$) rather than conv1’s weights directly, because BN’s normalisation step would otherwise absorb any raw weight scaling. The compensating $1/\alpha$ is applied to conv2’s input-channel dimension. Since ReLU is 1-homogeneous, the net effect on conv2’s input cancels and the block’s output is unchanged for any $\alpha > 0$. `verify_reparam_resnet` confirms a max absolute output deviation of $< 10^{-5}$ across all tested α values.

ViT-B/32 experiments are currently paused: GELU is not exactly 1-homogeneous, so the reparametrization via simple weight scaling introduces a measurable output deviation (~ 0.06 per unit for $\alpha = 2$). This issue is being investigated; a corrected formulation will be incorporated before the final report.

4 Next Steps

1. **Reparametrization invariance experiment (Experiment 2).** Run the full sweep: for each of the four optimizers, apply the five scaling factors $\alpha \in \{0.1, 0.5, 1.0, 2.0, 10.0\}$ to the ResNet-18 initialization and train for 200 epochs. Report per-optimizer variance of final test accuracy across α values and test the hypothesis that SAM is most sensitive, ASAM is partially invariant, and M-SAM is most invariant.
2. **Multi-seed evaluation.** Re-run the best configurations (SAM $\rho=0.02$, ASAM $\rho=0.5$, M-SAM $\rho=0.05$, SGD) with seeds $\{0, 1, 2\}$ (in addition to seed 42) to obtain SEM estimates and statistically sound comparisons. Current single-seed results show consistent trends but cannot support significance claims.
3. **Sharpness-generalization correlation.** With multi-seed data, compute the Pearson correlation between $\text{tr}(H)/d$ and divergence rate Δ across all optimizer- ρ combinations to quantify how well local sharpness predicts generalization.
4. **ViT experiments (stretch goal).** Resolve the GELU reparametrization approximation and replicate the key experiments on ViT-B/32 to test whether geometry-aware invariance generalises across architectures.

5 Team Contributions

- **Manav Dahra:** End-to-end implementation of all four optimizers (SGD, SAM, ASAM, M-SAM); ResNet-18 and ViT-B/32 model adapters; CIFAR-10 data pipeline; training loop with two-step SAM closure support; reparametrization utility for ResNet-18 BasicBlocks; Hutchinson trace estimator and loss landscape analysis; experiment runner scripts and YAML configuration system; baseline training sweep (12 checkpoint runs); initial results collection and this milestone writeup.

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A Loss Landscapes — All Checkpoints

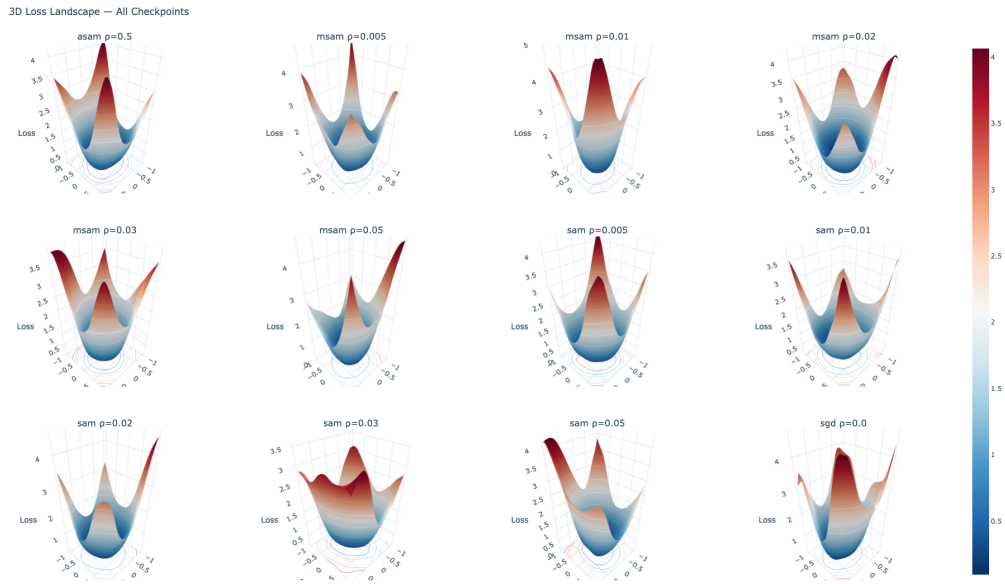


Figure 3: 3D filter-normalized loss landscapes for all 12 checkpoints (SGD, SAM $\times 5$, ASAM, M-SAM $\times 5$) on ResNet-18/CIFAR-10. Directions are Gram-Schmidt-orthogonalized filter-normalized random vectors; grid is 31×31 over $\alpha, \beta \in [-1, 1]$, clipped at loss = 5.0. Color encodes loss (blue = low, red = high).